

RSC Class 3

Elasticities, Marginal Revenue, and Costs **Market Analysis: Monopoly**

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Where we left off last time

- Analyzed markets using some algebra
- Defined elasticities as a concept to quantify economic behavior

The goal for today is to further operationalize the supply and demand model for analyzing competition in markets

Wrapping up Elasticities, and Marginal Revenue

Elasticity

We left off here...

Def. (Own-price) Elasticity is the ratio of percentage change in quantity to the percentage change in price.

We define the **(own price) elasticity of demand** ε as

$$\varepsilon = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q / Q}{\Delta P / P} = \frac{\Delta Q}{\Delta P} \frac{P}{Q}$$

$\Delta Q / \Delta P$ is slope of demand function we saw earlier

By definition, ε is negative, but we follow tradition and work with $|\varepsilon|$ as the elasticity of demand

Elasticity

We left off here...

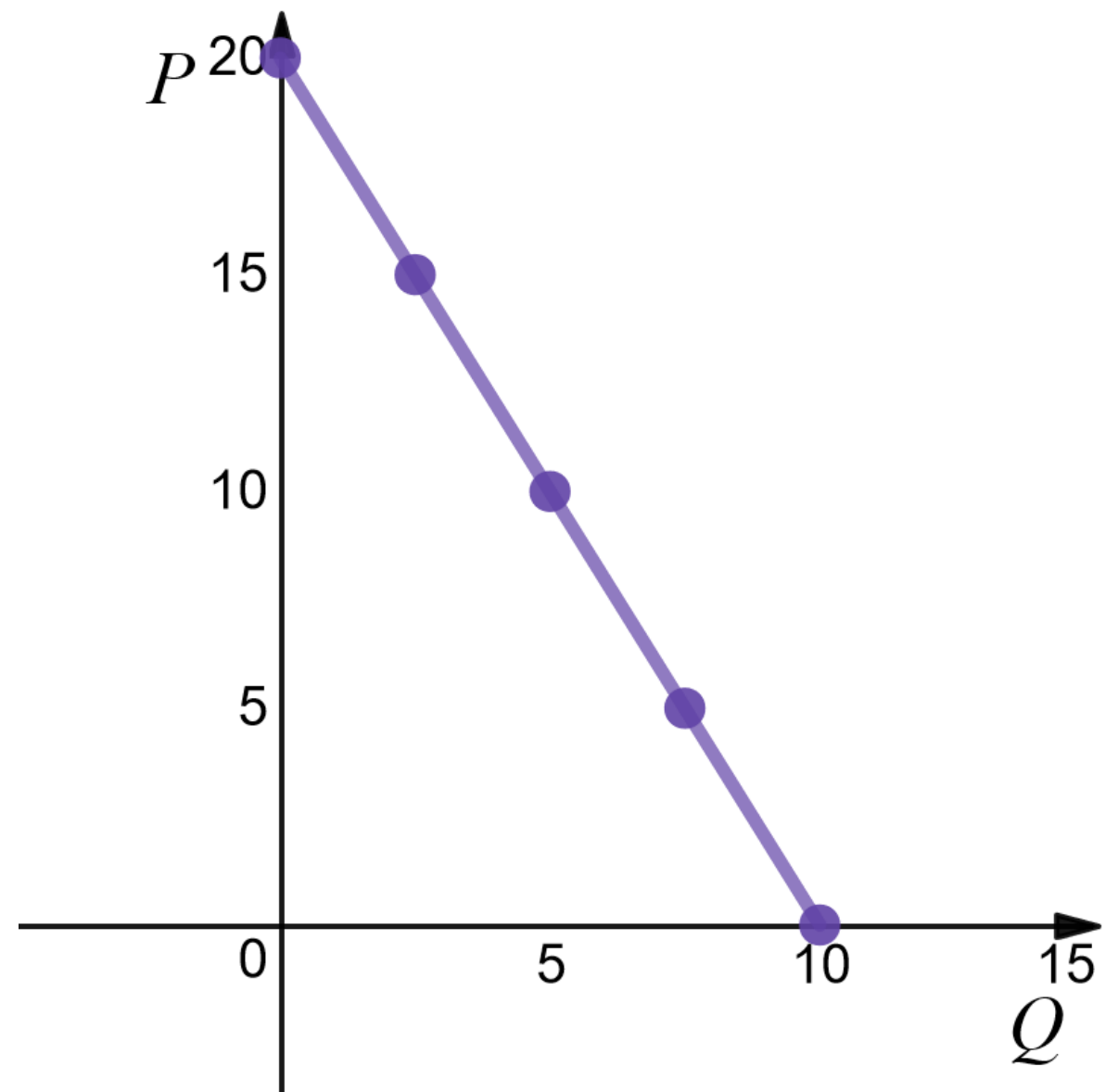
In addition to (own-price) elasticity of demand, i.e. point elasticities, we can also say demand is elastic or inelastic to mean something else

Def. Demand is (relatively) elastic if, for a given percentage change in price, the resulting percentage change in quantity demanded is large.

Def. Demand is (relatively) inelastic if, for a given percentage change in price, the resulting percentage change in quantity demanded is small.

Elasticities with linear demand

- Elastic above midpoint
- Unit-elastic at midpoint
- Inelastic below midpoint



Elasticity

Other elasticity measures

A few more pieces of terminology

Def. For any two goods X and Y , if the cross-price elasticity of demand for X with respect to Y is *negative (positive)*, then we say X and Y are *complements (substitutes)*.

Def. For any good X , if the income elasticity of demand for X is *negative (positive)*, then we say X is *inferior (normal)* good.

It is important to know that we are *never* told the exact elasticities of demand in life; we must infer them from context

WE WILL EXPLORE THE LAST POINT ABOVE IN PRACTICE QUESTIONS, ETC

Elasticity

Other elasticity measures

Examples.

1. Rex's net income increased by 21% this year compared to last; his purchases of sparkling water rose by 17%. What is his income elasticity of demand for sparkling water?

Elasticity

Other elasticity measures

Examples (continued).

2. Wicked Art Fair (WAF) and Artistic Odyssey (AO) hold fairs at the same time every year. But, this year, WAF raised their price by 20%, and AO sees their traffic increased by 11%. What is the cross-price elasticity of demand of (tickets for) AO?

Elasticity

With marginal revenue

We're now going to link elasticity to marginal revenue

The intellectual exercise is:

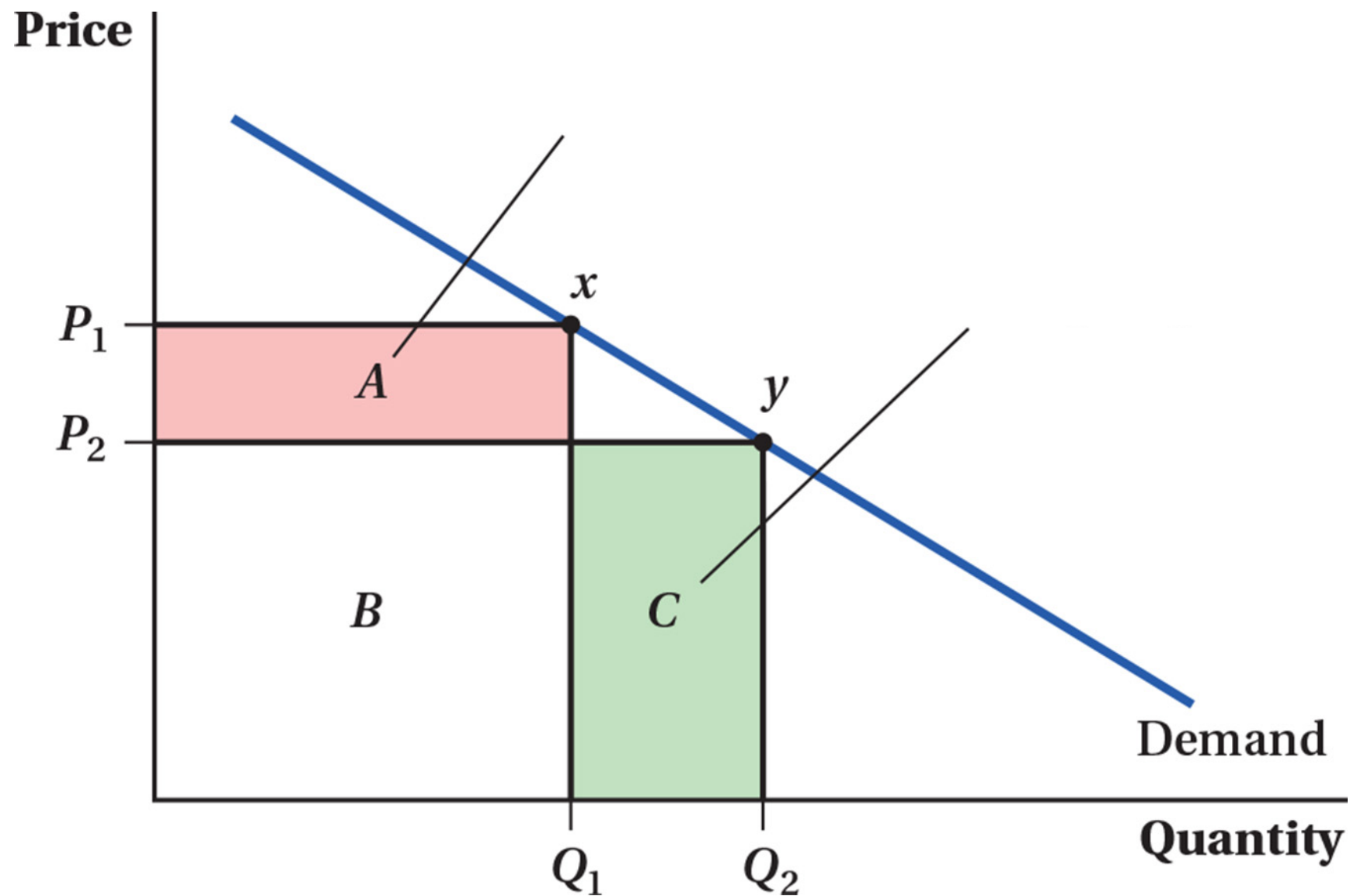
Imagine we own a firm, and we face a group of potential customers — each with the same demand for our product

How can I relate their behavior to incremental revenue we will earn as a firm?

You might ask: why should we, as a firm, care about this? And how would we even know the answer to this question?

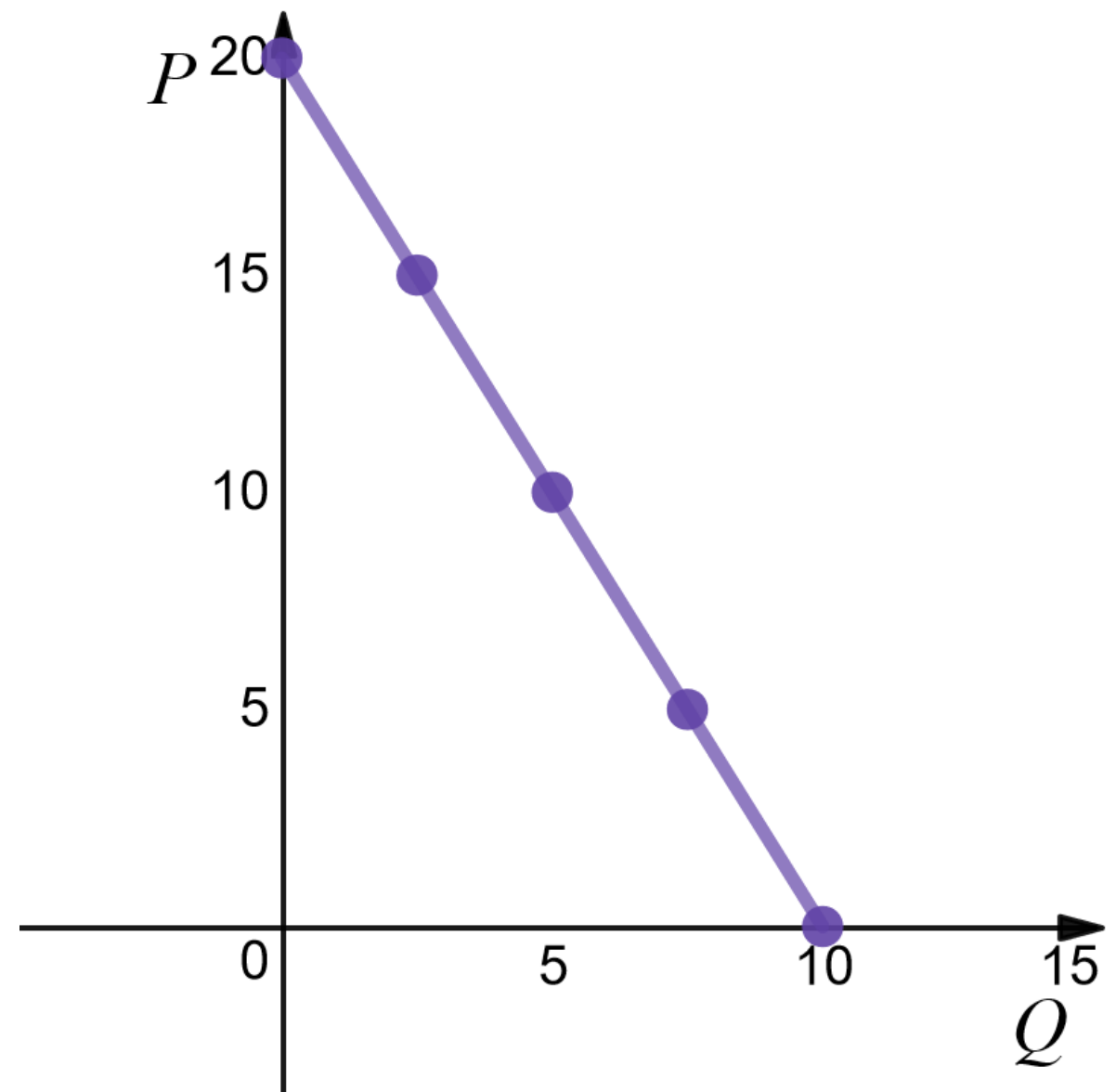
Elasticity

With marginal revenue



Elasticities with linear demand

- Elastic above midpoint
- Unit-elastic at midpoint
- Inelastic below midpoint



Fact: for a firm facing any given demand function, the **firm's marginal revenue is zero** when they set price = where the customer's demand is unit-elastic (at midpoint if we're working with linear demand)

This will our first foray into analyzing consumer behavior (and explaining why firms care about knowing elasticity)!

Elasticity

With marginal revenue

The fact from previous slide is a bit mysterious, let's unpack that

We will be using **marginal revenue** a lot going forward

This is the concept of marginal benefits for firms

Def. Marginal revenue is the incremental revenue made by firms when they produce one additional unit of product

Mathematically, we derive marginal revenue from the revenue function

We now discuss firms and introduce connections to elasticity of demand

Elasticity

With marginal revenue

First, a few facts (assumptions) in the supply/demand model:

Fact #1. Firms' objective is to maximize profits = total revenue minus total costs.

Fact #2. Firms choose quantity to maximize profits.

Fact #3 (!). Firms' demand (they face) may differ from the market demand for a product they sell.

Elasticity

With marginal revenue

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Fact #1. Firms' objective is to maximize profits = total revenue minus total costs.

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Fact #3 (!). Firms' demand (they face) may differ from the market demand for a product they sell.

Fact #1 + #2 $\implies$ Define **profit function** for a firm, in which the firm chooses **quantity**

$$\Pi(Q) = \text{revenue function}(Q) - \text{cost function}(Q)$$

Elasticity

With marginal revenue

Fact #1 + #2 $\implies$ Define **profit function** for a firm, in which the firm chooses **quantity**

$$\Pi(Q) = \text{revenue function}(Q) - \text{cost function}(Q)$$

Since we're thinking about firms choosing quantity, we cannot work with *demand functions* anymore: demand functions express quantity in terms of price

We need to work with **inverse demand functions**, which express price in terms of quantity

Def. A (linear) inverse demand function is of the form $P = a - bQ$.

Elasticity

With marginal revenue

Fact #1 + #2 $\implies$ Define **profit function** for a firm, in which the firm chooses **quantity**

$$\Pi(Q) = \text{revenue function}(Q) - \text{cost function}(Q)$$

Def. A (linear) inverse demand function is of the form $P = a - bQ$.

Revenue function, expressed in terms of firms choosing quantity to produce, is defined as

$$R(Q) = P(Q) \times Q$$

We will come back to costs, written as $C(Q)$, later today

Elasticity

With marginal revenue

Fact #1 + #2 $\implies$ Define **profit function** for a firm, in which the firm chooses **quantity**

$$\Pi(Q) = \text{revenue function}(Q) - \text{cost function}(Q)$$

Revenue function is $R(Q) = P(Q) \times Q$

Marginal revenue is the derivative of the revenue function $R(Q)$

With linear inverse demand function $P = a - bQ$,

$$R(Q) = (a - bQ)Q = aQ - bQ^2$$

Elasticity

With marginal revenue

Fact #1 + #2 $\implies$ Define **profit function** for a firm, in which the firm chooses **quantity**

$$\Pi(Q) = \text{revenue function}(Q) - \text{cost function}(Q)$$

Revenue function is $R(Q) = P(Q) \times Q$

Applying the power rule to $R(Q) = (a - bQ)Q = aQ - bQ^2$,

$$MR(Q) = a - 2bQ$$

is the expression for marginal revenue function

Elasticity

With marginal revenue

Fact #1 + #2 $\implies$ Define **profit function** for a firm, in which the firm chooses **quantity**

$$\Pi(Q) = \text{revenue function}(Q) - \text{cost function}(Q)$$

Fact. With linear inverse demand function $P = a - bQ$, the marginal revenue function associated with the demand is

$$MR(Q) = a - 2bQ$$

i.e. shares the same intercept as the linear inverse demand function, but has twice the slope!

Important note. Use the inverse demand function to use this fact to derive the marginal cost function!

Elasticity

With marginal revenue

Now, let's relate marginal revenue function to elasticity of demand where the demand is linear

We need to do a bit of algebra, writing ε as shorthand for ε_D ...

$$\varepsilon = \frac{\Delta Q}{\Delta P} \frac{P}{Q} \iff \frac{Q}{P} \varepsilon = \frac{\Delta Q}{\Delta P} \iff \overbrace{\frac{P}{Q} \frac{1}{\varepsilon}}^{\text{holds for linear demand function}} = \frac{\Delta P}{\Delta Q}$$

and $MR(Q) = P'(Q)Q + P(Q)$ (from earlier)

$$\implies MR(Q) = \frac{P(Q)}{Q} \frac{1}{\varepsilon} Q + P(Q) = P(Q) \left(1 + \frac{1}{\varepsilon} \right)$$

Elasticity

With marginal revenue

Now, let's relate marginal revenue function to elasticity of demand where the demand is linear

$$MR(Q) = P(Q) \left(1 + \frac{1}{\varepsilon} \right)$$

Notice that $P(Q)$ is the inverse demand function

This means marginal revenue is *increasing* in ε (because ε — the own-price/point elasticity of demand — is negative)!

This means: when consumers are sensitive to price changes, firms have the incentive to lower prices to earn (more!) positive marginal revenue!

Elasticity

With marginal revenue

So, two things: with a linear inverse demand $P = a - bQ$,

$$(\star) \quad MR(Q) = a - 2bQ, \text{ and}$$

$$(\star\star) \quad MR(Q) = P(Q) \left(1 + \frac{1}{\varepsilon} \right)$$

It is way more natural to plot $(\star)$, and use $(\star\star)$ to argue the **Fact** from earlier

Graphically, we have

Elasticity

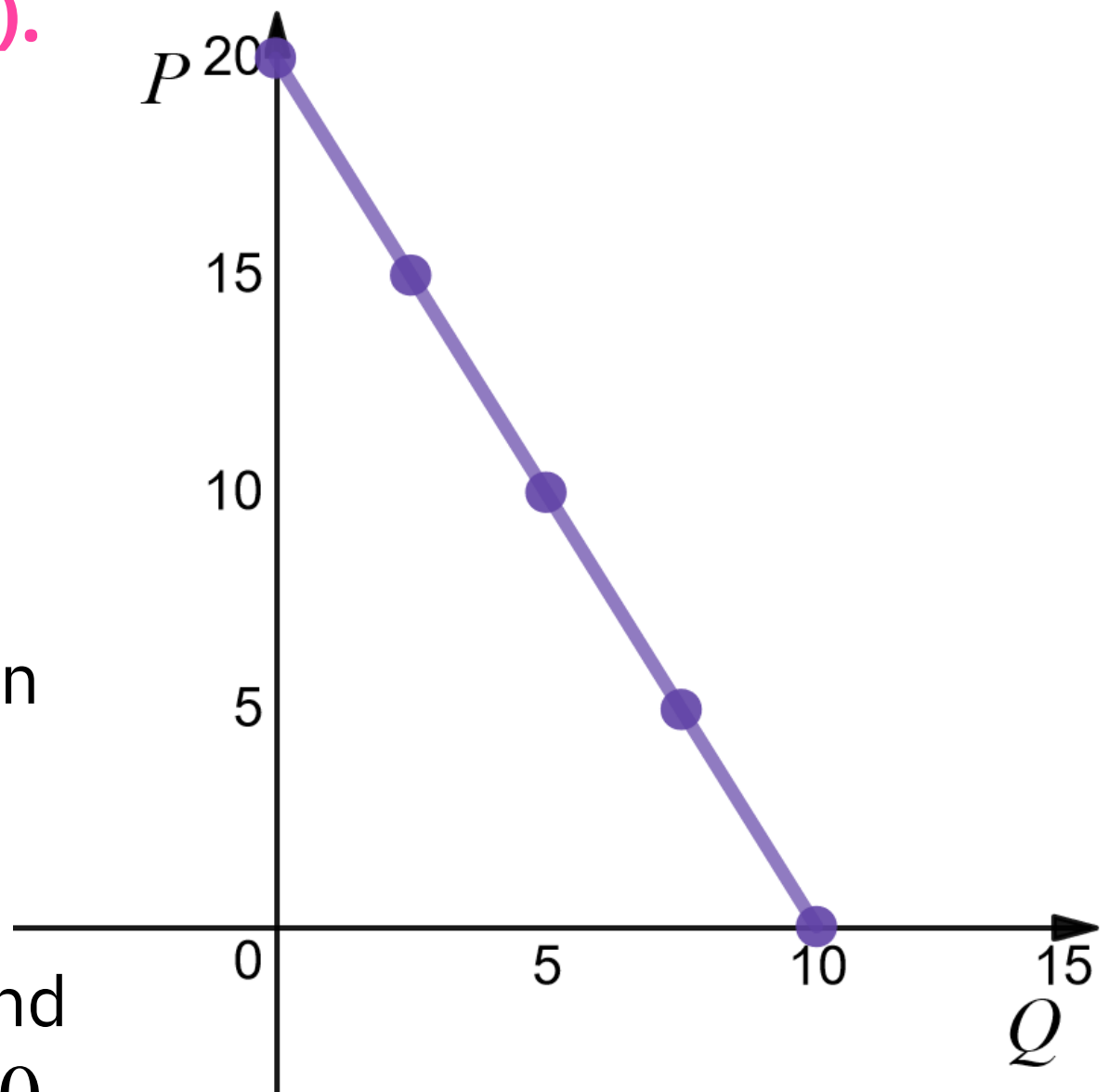
With marginal revenue

Example (Continued from Class #2).

Given the demand function

$$Q = 10 - 0.5P,$$

1. Calculate the inverse demand function and marginal revenue function.
2. Plot the marginal revenue function alongside the inverse demand function.
3. Calculate the marginal revenue and elasticity of demand when $P = 10$.



Elasticity

With marginal revenue

So, two things: with a linear inverse demand $P = P(Q) = a - bQ$,

$$(\star) \quad MR(Q) = a - 2bQ, \text{ and}$$

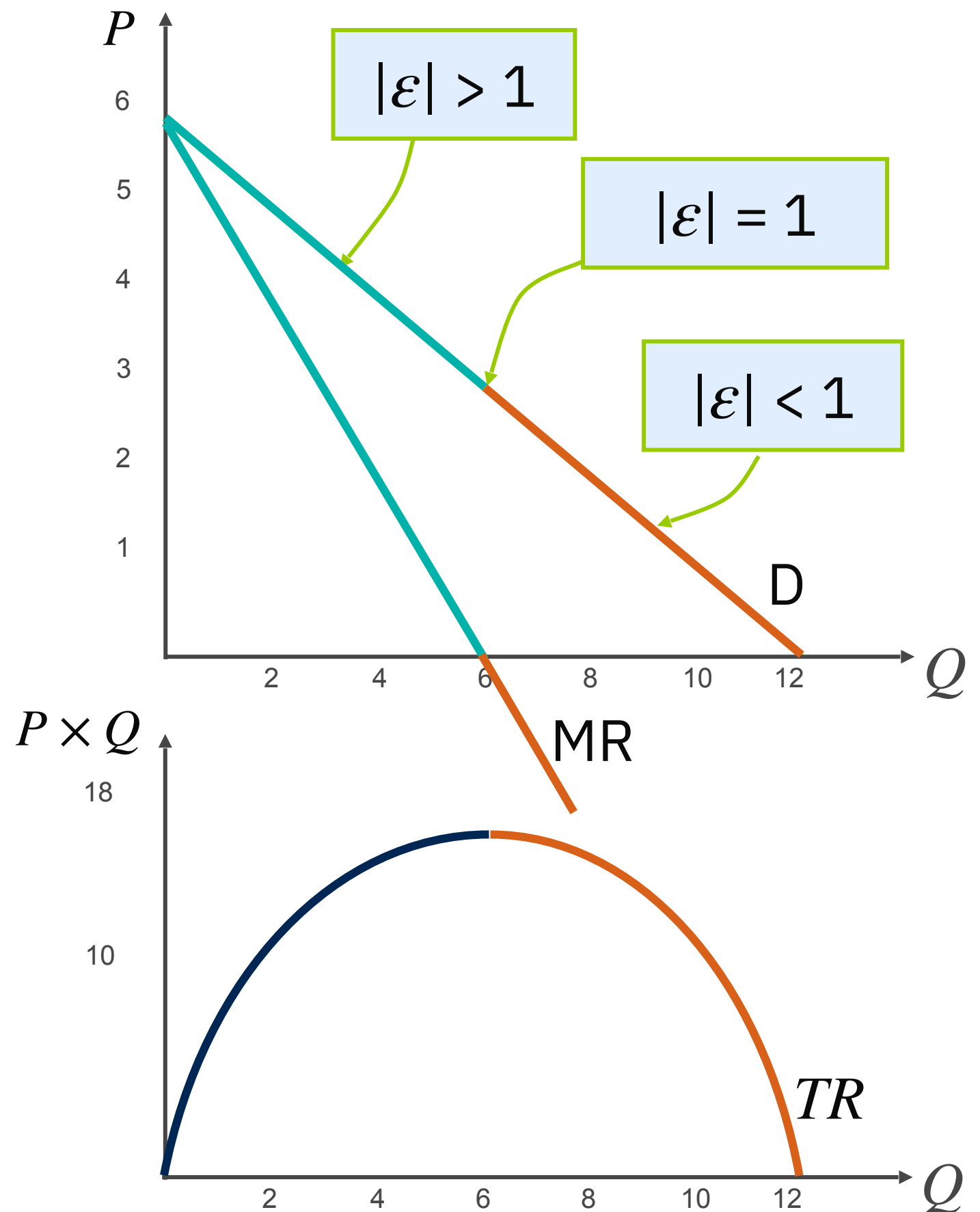
$$(\star\star) \quad MR(Q) = P(Q) \left(1 + \frac{1}{\varepsilon} \right)$$

Final fact (connect the dots!): calculus tells us that **revenue is maximized when marginal revenue = 0**, so in the case of linear demand...

Elasticity

With marginal revenue

To summarize...



Elasticity

With marginal revenue

Now, remember Fact #1 and Fact #2:

Fact #1. Firms' objective is to maximize profits = total revenue minus total costs.

Fact #2. Firms choose quantity to maximize profits — not revenues!

Firms maximize profits Π , i.e.

$$\Pi = \text{Total Revenue} - \text{Total Costs}$$

This means we need to think about firms' costs!

Costs

Costs

We're now talking about firms

We've talked about costs for individuals as they make decisions already (sunk costs, opportunity costs)

In economics, firms are a “black box” producing goods and services

- Producing anything requires resources $\Leftrightarrow$ trade-offs; we capture trade-offs in terms of cost functions

Def. The total cost function $C(Q)$ captures the sum of all *economic* costs for a firm producing at quantity Q . Sometimes, total cost function is also written as $TC(Q)$

How many types of costs are there?

Costs

We're now talking about firms

How many types of costs are there?

We talked about opportunity costs in Class #1

There are two types of *production (firm's)* costs we care about

Def. Fixed costs are costs that do not change with the level of quantity of produced

Def. Variable costs are costs that vary with the level of quantity produced

Costs

We're now talking about firms

How many types of costs are there?

We write total cost $C(Q) = FC + VC(Q)$

FC is a constant number

$VC(Q)$ is a function of quantity

Marginal cost $MC(Q)$ is the derivative of total cost $C(Q)$

Costs

We're now talking about firms

How many types of costs are there?

We write total cost $C(Q) = FC + VC(Q)$

FC is a constant number, and $VC(Q)$ is a function of quantity

Marginal cost $MC(Q)$ is the derivative of total cost $C(Q)$

Combined, profit function is

$$\Pi(Q) = R(Q) - C(Q) = P(Q)Q - C(Q)$$

Costs

We're now talking about firms

How many types of costs are there?

We write total cost $C(Q) = FC + VC(Q)$

FC is a constant number, and $VC(Q)$ is a function of quantity

We can also talk about average fixed cost (function) and average variable cost (function), where the average is taken over quantities

Def. Average total cost $ATC(Q) = C(Q)/Q$

Def. Average fixed cost $AFC(Q) = FC/Q$

Def. Average variable cost $AVC(Q) = VC(Q)/Q$

Costs

We're now talking about firms

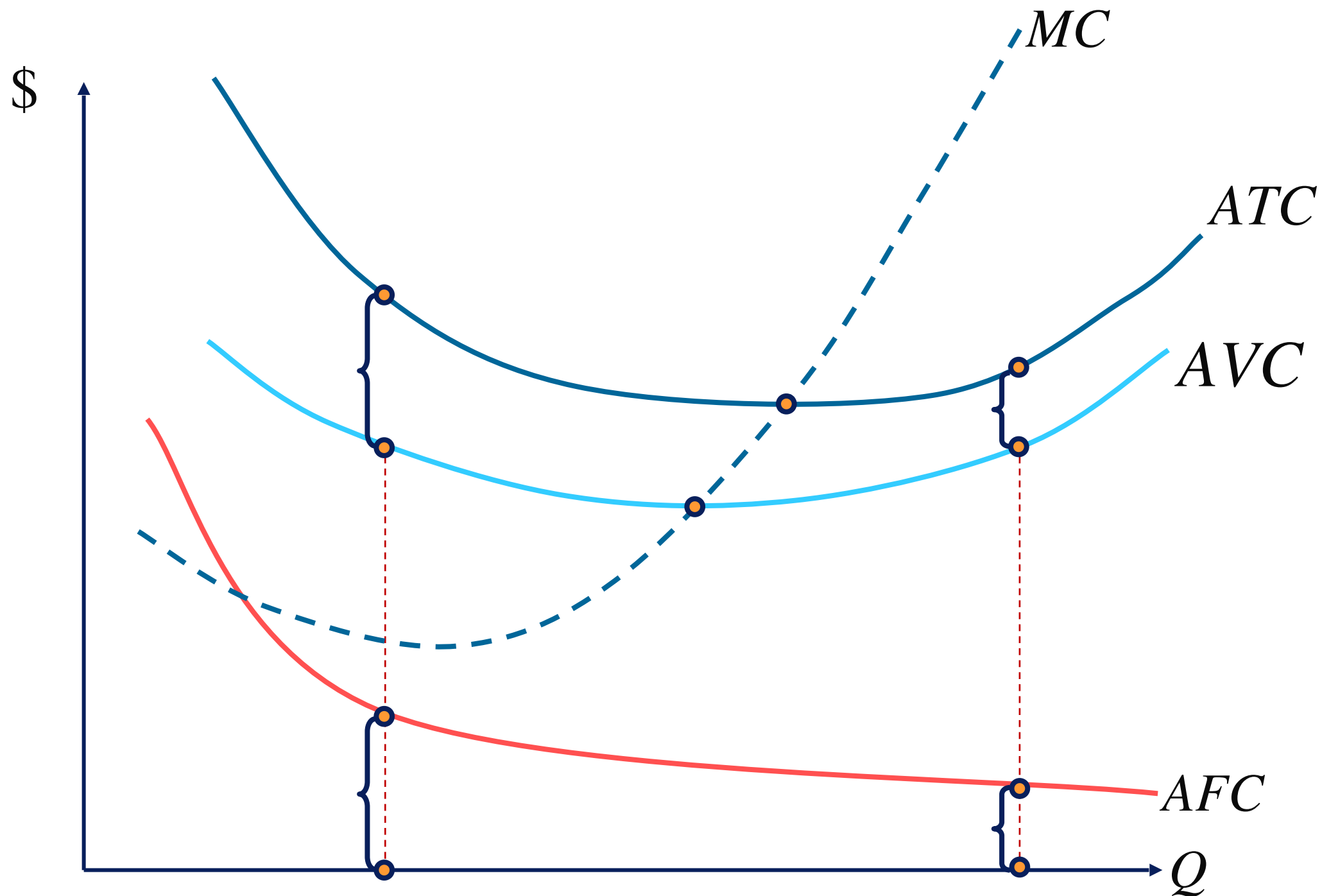
How many types of costs are there?

Example. Given total cost function $C(Q) = 100 + 24Q - 8Q^2 + Q^3$,

- What is TFC?
- What is TVC?
- What is ATC?
- What is AFC?
- What is AVC?
- Marginal cost?

Costs

We're now talking about firms



Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Remember that firms maximize profits by setting the quantity according to the fundamental decision rule, which is:

_____ equal _____

Thinking on the margin $\Rightarrow$ any cost incurred in the past does not matter

Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Def. Costs that occurred in the past, and are irrecoverable, are **sunk costs**.

Sunk costs do not matter from the point of view of decision-making

Thinking that they do is called sunk cost fallacy

Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Example. Rex is a licensed headphone reseller and owns a shop. He bought 1,000 units of the latest Sony wireless headphones for \$500/unit.

Every pair of headphone purchased requires an initial maintenance, costing \$160/unit per year. This is an unavoidable cost.

Right now, he received an offer from another reseller, who offers to buy all 1,000 units from him for \$450/unit. Should he sell all of his stock or not?

Costs

We're now talking about firms

What costs *matter* for firms' decision-making process?

Example. Rex is a licensed headphone reseller and owns a shop. He bought 1,000 units of the latest Sony wireless headphones for \$500/unit.

To run the shop, he hires 18 clerks at \$55,000/year; pays \$30,000 in utilities per year; pays himself \$270,000/year.

Every pair of headphone purchased requires an initial maintenance, costing \$160/unit per year. This is an unavoidable cost.

He sold all 1,000 headphones. He's got a buyer for another headphone (of the same model). What is his marginal cost of selling 1 additional unit of the same headphone model?

Costs

We're now talking about firms

We've nailed down consumer behavior, firm's marginal revenue, and firm's costs — separately

We are now ready to put all of these together and look at a market, with supply and demand, together

The simplest form of competition: there is none!

This is called monopoly

For next time

Next time: market analysis!

Come to office hours w/ any questions/concerns

See you tomorrow (Wednesday @ 9am sharp)!